

Rare Earths: The Vitamins Of Industry

[B2]

Essenziali per la produzione di ogni tipo di tecnologia, dai telefoni cellulari agli armamenti, questi materiali sono diventati una delle risorse naturali più preziose e ambite al mondo.



Every time you use your smartphone or drive an electric vehicle, you benefit from technological advancements made possible by rare-earth elements (REEs, or '[rare earths](#)'). Also called "the vitamins of modern industry," a phrase attributed to Soviet scientist Aleksandr Fersman, [rare earths](#) have become one of nature's most valuable [resources](#). This is because of [soaring](#) demand for advanced electronics, renewable energy technology, and military and defence systems. All of these [harness](#) the unique magnetic, optical, and catalytic properties of these critical [minerals](#).

SOLAR PANELS AND MISSILES

So what are [rare earths](#)? They are seventeen metallic elements: fifteen lanthanides, a closely related series of metals [running from lanthanum to lutetium](#), known for their similar chemical behaviour, and two soft, [silvery metals](#) called Scandium and Yttrium. They are often divided into light [rare earths](#) and heavy [rare earths](#). However, they all have almost identical chemical and physical properties, and are typically found in the same [mineral](#) deposits. This makes them extremely difficult to separate in their [raw](#), natural form. Rare earths are used in the development of powerful magnets and are essential for modern technologies including consumer electronics, such as laptops or [headsets](#); green energy applications, including solar panels and wind turbines; medical devices such as X-ray equipment and lasers; and military machines, such as missiles and [fighter jets](#).

NOT THAT RARE

The surprising thing about [rare earths](#) is that they are not, in fact, [scarce](#). They are found worldwide in the [Earth's crust](#). However, because of the difficulty in separating them from each other, mining and refining them is extremely costly and complex. These extractive processes cause significant environmental damage, including habitat destruction, [waste](#) generation and pollution. The race for REEs have [heightened tensions](#) between China and the US, provided Russia with an incentive for invading Ukraine and have impacted Latin American countries, such as Colombia (above) and Venezuela.

CHINESE DOMINATION

An accurate predictor of future trends, China began investing in complex technologies to [mine](#), separate and process [rare earths](#) in the 1980s. That is why today the country is the global leader in [rare earths](#), mining up to 70 per cent and processing and refining about 90 per cent of the world's [supply](#). Naturally, other countries are trying to compete, by developing their own processing capabilities. And while China continues to dominate the market, the US, Myanmar and Australia have become major producers in recent years. The global market value

for products containing [rare earths](#) is well over \$1 [trillion](#). And despite the financial and environmental costs involved in their production, the demand for [rare earths](#) is expected to grow exponentially over the next few years as our dependence on technology, and AI in particular, becomes ever more acute. Culture

Bennu: Recipe For Life La NASA sta studiando i campioni prelevati da questo asteroide dal nome egizio: contengono importanti informazioni sulla formazione del sistema solare e sull'origine della vita sulla Terra. Daniel Francis , Talitha Linehan

GEOPOLITICS

Rare earths have long been a geopolitical [concern](#). They are behind aggressive actions, such as the Russian invasion of Ukraine, and the US push to obtain Greenland. Both Ukraine and Greenland are home to significant rare earth deposits. US rhetoric regarding acquisition of [mineral](#) rights has also led to [heightened tensions](#) with Venezuela. Interest in the country is [tied](#) to its oil sector and also to its [broader mineral](#) wealth. [Reported occurrences](#) of critical [minerals](#) include [rare earth-bearing sands](#) in the Orinoco Mining Arc. Because [rare earths](#) are critical for magnets, sensors, lasers and guidance systems, the country that controls separation and processing [wields](#) considerable economic power. China's state-subsidised investment in solvent extraction and [midstream](#) refining (highly-polluting processes) has created a [bottleneck](#); as China processes the vast majority of the world's REEs, it can restrict [supply](#) to [leverage](#) its advantage.

Glossary

- **soaring** = crescente
- **headsets** = auricolari, cuffie
- **waste** = rifiuti
- **mine** = estrarre
- **supply** = fornitura
- **broadier** = più ampia
- **Reported occurrences** = giacimenti segnalati
- **leverage** = sfruttare
- **harness** = sfruttare
- **silvery metals** = metalli argentati
- **fighter jets** = aerei da combattimento
- **Earth's crust** = crosta terrestre
- **heightened tensions** = forti tensioni
- **rare earth-bearing sands** = sabbie contenenti terre rare
- **bottleneck** = punto critico, ostacolo
- **rare earths** = terre rare
- **resources** = risorse
- **running from lanthanum to** = che vanno da... a...
- **tied** = legare
- **midstream** = nella fase intermedia
- **raw** = allo stato grezzo
- **scarce** = scarse, poco diffuse
- **trillion** = mille miliardi
- **concern** = preoccupazione
- **wields** = esercitare